

Name: _____

Date: _____

TRIGONOMETRY, (SOH CAH TOA FINDING ANGLES)

LO: I can use inverse trigonometric functions to find unknown angles in right-angled triangles.

Finding angles with trigonometry

To find an **unknown angle**, use the inverse trig functions: $\sin^{-1}$, $\cos^{-1}$, or $\tan^{-1}$. Choose which ratio to use based on which **two sides** are known.

SOH CAH TOA: Sin = Opposite/Hypotenuse, Cos = Adjacent/Hypotenuse, Tan = Opposite/Adjacent.

Quick examples

●	opp = 3, hyp = 6: angle = $\sin^{-1}(\frac{3}{6}) = 30^\circ$	use SOH
●	adj = 4, hyp = 8: angle = $\cos^{-1}(\frac{4}{8}) = 60^\circ$	use CAH
●	opp = 5, adj = 5: angle = $\tan^{-1}(1) = 45^\circ$	use TOA
●	Use sin when you have adjacent and hypotenuse	that is cos (CAH)

Key idea

Identify the two known sides, choose the correct ratio, then apply the inverse function.

$$\theta = \sin^{-1}(\text{opp} \div \text{hyp})$$

Or $\cos^{-1}(\text{adj/hyp})$ or $\tan^{-1}(\text{opp/adj})$

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - using sin inverse

Find angle x: opposite = 4, hypotenuse = 8.

Have opposite and hypotenuse: use

sin

$$\sin x = \frac{4}{8} = 0.5$$

$$x = \sin^{-1}(0.5) = 30^\circ$$

$$x = 30^\circ$$

Opposite and hypotenuse means SOH.

Example 2 - using cos inverse

Find angle x: adjacent = 5, hypotenuse = 10.

Have adjacent and hypotenuse: use

cos

$$\cos x = \frac{5}{10} = 0.5$$

$$x = \cos^{-1}(0.5) = 60^\circ$$

$$x = 60^\circ$$

Adjacent and hypotenuse means CAH.

Example 3 - using tan inverse

Find angle x: opposite = 7, adjacent = 7.

Have opposite and adjacent: use

tan

$$\tan x = \frac{7}{7} = 1$$

$$x = \tan^{-1}(1) = 45^\circ$$

$$x = 45^\circ$$

Opposite and adjacent means TOA.

Example 4 - decimal answer

Find angle x: opposite = 6, hypotenuse = 9.

$$\sin x = \frac{6}{9} = 0.667...$$

$$x = \sin^{-1}(0.667)$$

$$x = 41.8^\circ \text{ (1 dp)}$$

$$x = 41.8^\circ$$

Round to 1 decimal place unless told otherwise.

YOUR TURN – PRACTICE (deliberate practice)

1. Using $\sin^{-1}$

Find each angle. Give exact values where possible.

a) opp = 5, hyp = 10

b) opp = 3, hyp = 6

c) opp = 4, hyp = 4

d) opp = 6, hyp = 12

2. Using $\cos^{-1}$

Find each angle.

a) adj = 3, hyp = 6

b) adj = 5, hyp = 10

c) adj = 7, hyp = 14

d) adj = 4, hyp = 4

3. Using $\tan^{-1}$

Find each angle.

a) opp = 8, adj = 8

b) opp = 5, adj = 3

c) opp = 3, adj = 3

d) opp = 10, adj = 10

4. Choosing the correct ratio

Identify which ratio to use, then find the angle (1 dp).

a) opp = 7, hyp = 11

b) adj = 6, hyp = 9

c) opp = 8, adj = 5

d) adj = 12, hyp = 15

5. Mixed angle problems

Find angle x in each triangle (1 dp where needed).

a) opp = 4, adj = 3

b) opp = 5, hyp = 13

c) adj = 8, hyp = 10

d) opp = 12, adj = 9

e) opp = 1, hyp = 2

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) To find an angle, use the inverse trig function ☐

Correct answer: _____

Reason: _____

b) $\sin^{-1}(0.5) = 60^\circ$ ☐

Correct answer: _____

Reason: _____

c) $\tan^{-1}(1) = 45^\circ$ ☐

Correct answer: _____

Reason: _____

d) Use SOH when you know the adjacent and hypotenuse ☐

Correct answer: _____

Reason: _____

e) $\cos^{-1}(0.5) = 60^\circ$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A ladder **6 m long** leans against a wall. The foot is **2.5 m** from the base of the wall. Find the angle between the ladder and the ground. If the angle must be less than **75°** for safety, is this ladder safe?

Expression: _____

Simplified: _____

b) From a point **30 m** from the base of a tower, the angle of elevation to the top is **55°**. Find the height of the tower. A second person stands where the angle of elevation is **35°**. How far are they from the base?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

